Design Blueprint

A: **Problem Context and Project Summary**

Ocular diseases are an issue that are responsible for many cases of preventable blindness worldwide, with many people at risk of developing these diseases needing monitoring to ensure early treatment occurs to prevent the worst case scenario. Currently, deep learning research in the field focuses on supervised approaches with large datasets needed models often blind to diseases out of their focus. This project aims to solve this through the development of a convolutional auto encoder trained on exclusively healthy cases for unsupervised anomaly detection. Images not reconstructed with high accuracy can therefore be flagged at risk for having potential anomalies worth investigating without having to specifically be trained on certain diseases like glaucoma.

B: **Dataset**

The dataset used for the project is the ODIR-5k. Accessed through Kaggle, it is a dataset of 5000 patients with images from their left and right eyes and diagnostics from their doctors. The data was collect collected by Shanggong Medical Technology Co., Ltd. from different hospitals/medical centers in China, and labeled by trained humans for 6 specific diseases, a class for other diseases and anomalies, and a normal label. There are over 6,000 labeled images, with 2,101 strictly normal patients, meaning that both of their eyes are normal with no diseases/anomalies, which we be used for training purposes. The images are JPEGs, with some image size normalization needed. There is also a separate CSV of other data like age for every patient.

Preprocessing challenges include: resizing all images to 256x256, augmentation of the images to make vessels more visible similar to other augmented versions of this dataset, border padding to prevent stuff outside eye impacting, and standard flips, rotations, and color swapping to augment data size of the strictly normal set.

The dataset was collected under Chinese privacy regulations. This does mean sample size is limited to Chinese population and may have some generalization limitations to other populations, and images collected with different equipment than used in the study may have different results depending on how the imaging captures retina.

B: **Planned Architecture**

Currently, the plan is to develop a convolution autoencoder in pytorch. Four convolution blocks with a Conv2D -> BatchNorm _> Relu (possibly leaky) -> MaxPool2D structure. The 256x256 images will be compressed down to 16x16 with increasing channel depth each time. The decoder will mirror this structure to reconstruct the image. Two loss functions will be used: MSE and SSIM with combination of the two likely producing the best result

Data
Centercrop, resize, normalize, CLAHE (color channel filter)

prepocessed tensor
- 3x256x256

Encoder
- Conv2D blocks + MaxPool -> 128x16x16 bottleneck

Decoder
- TransposeConv Blocks -> 3x256x256 reconstructed image

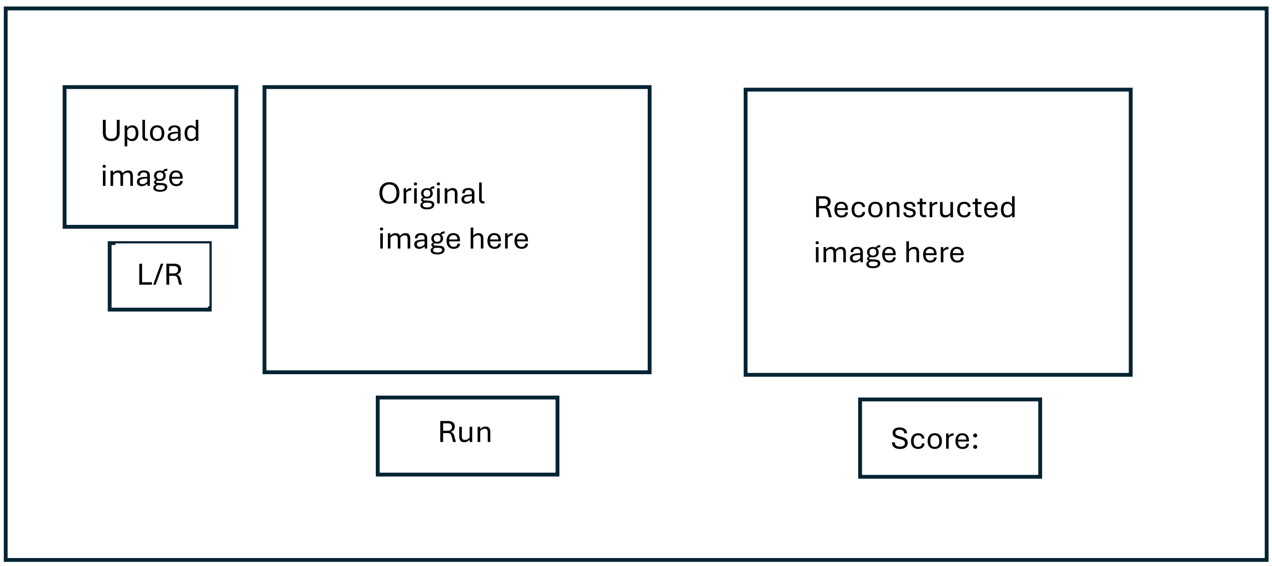
Reconstruct Error
- MSE+SSIM per pixel for anomaly score

UI
- Gradio UI with normal/anomaly verdict, possible heatmap if time permits

Computation will be done on Hypergator or Google collab, local if depending on performance. Gradio interface will be built in order to create easier experience for users to upload images and get anomaly feedback.

D: **UI plan**

User will upload retinal image, with left/right flagging option provided but not required. The model connected to gradio will reconstruct the image, with the reconstruction shown next to the original image. An anomaly score from 0-1 will be provided based on loss threshold determined from training. If time permits, a heatmap will be displayed over reconstructed image showing where anomaly locations may be. This enhances user experience by providing both a numerical score to indicate how likely there is an anomaly, as well as showing visual locations on the reconstructed image, which is displayed next to the original for reference to make it clearer where the potential issue may be.



E: **Innovation and Anticipated Challenges**

- Training on strictly healthy images (Cases where both eyes have no diseases or anomalies) means zero disease annotations are needed and model is much easier to deploy.
- Thid does mean the autoencoder is evaluated against 7 unseen disease categories, and can have a per class AUROC for each, generalizing for all of them from a single class training.
- One risk is limited normal labeling, this can be countered through data augmentation, possible inclusion of other dataset images, and in worst case dropping strictly normal constraints and allow training on cases where the other eye has an anomaly flagged.
- Reconstruction threshold may be sensitive, and will require careful finetuning.

- Well thought preprocessing will need to be conducted to ensure quality remains consistent of training images and that vessels are easier for the model to see.

F: **Implementation Timeline**

- Week 4/5-4/12: Basic model implementation, with possible per class checks
- Week 4/12-4/19: Model final tuning, UI development in Gradio
- Week 4/19-4/26: Final touches/buffer week, reported started
- Week 4/26-29: Report finished

G: **Responsible AI Reflection**

As mentioned previously, model was exclusively trained on dataset from China, so generalization across other populations may differ. Same applies to imaging collected through other equipment then put through the trained model. UI design aims to provide transparency through visual and numerical metrics of anomaly detection, with documentation about any thresholds and justification within project report. No environmental mental concerns are expected due to low cost of running.